



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO ANNAI FUEL SERVICE

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

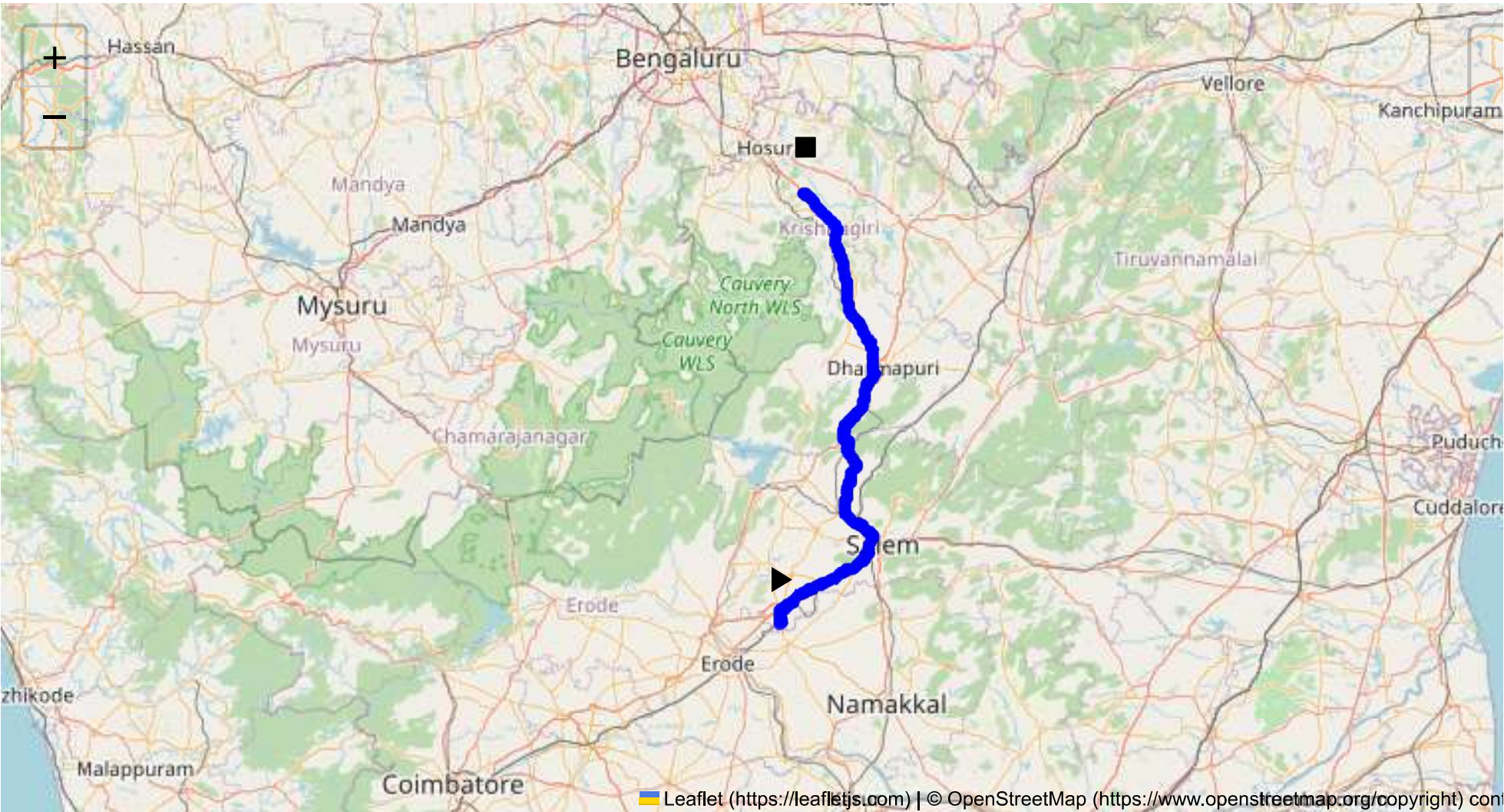
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 165.52 km**  
**Estimated Duration: 3.1 hours**  
**Adjusted Duration (Heavy Vehicle): 3.9 hours**  
**Start: (11.4381, 77.8734)**  
**End: (12.611051, 77.935165)**



# Welcome to the Journey Risk Management Study

## Route Overview

The route from CVQF+23W, Sangagiri, Tamil Nadu 637301, India to JW6P+84Q, Uddanapalli, Tamil Nadu 635119, India covers approximately 165.52 kilometers. This route likely follows national highways and state roads commonly used for both local and regional transportation needs. The journey typically takes about 3.10 hours for heavy vehicles carrying hazardous materials.



## Weather Conditions

- **Typical Weather:** The region experiences a tropical climate with hot summers (March to June), monsoons (June to September), and mild winters (November to February).
- **Weather Hazards:** Monsoon season may bring heavy rains, leading to waterlogging and potential flooding in low-lying areas. Drivers should be cautious of slippery roads and reduced visibility during heavy rainfall.

## Traffic Patterns

- **Peak Hours:** Urban areas near the start and end points may experience congestion during typical commuting hours (8:00 AM - 10:00 AM and 5:00 PM - 8:00 PM).
- **Congestion-Prone Areas:** Expect traffic slowdowns near larger towns and cities, including Erode and Krishnagiri, especially at junctions with busy local roads.

## Road Quality and Infrastructure

- **Road Quality:** Main highways are generally well-maintained, but secondary roads may have potholes, especially after the rainy season. Some rural roads might be narrower and require careful navigation.
- **Infrastructure:** Adequate infrastructure exists with petrol stations and rest stops along the highway. However, services may become sparse in more rural segments.

## Alternative Routes for Emergencies

- Consider NH44 as an alternative for portions of the journey to bypass potentially hazardous or congested areas. It's important to have updated maps for real-time detours in case of unexpected closures or traffic.

## Local Regulations for Hazardous Material Transport

- **Regulations:** Heavy vehicles must adhere to local and national transport regulations, including obtaining necessary permits for hazardous materials. Restrictions on driving through densely populated areas during peak hours might apply.
- **Requirements:** Drivers should be familiar with documentation requirements and carry material safety data sheets (MSDS) for all hazardous materials.

## Historical Incidents

- **Past Incidents:** Highways in Tamil Nadu have seen accidents involving heavy vehicles, often due to speeding, driver fatigue, or poor road conditions. It is crucial to maintain safe speeds and vigilance at all times.

## Environmental Considerations

- **Sensitive Areas:** Sections of the route may pass near agricultural lands and water bodies, requiring extra care to avoid contamination risks from hazardous materials.
- **Wildlife Crossings:** Be alert for animal crossings in rural and forested areas, particularly in the evening and night.

## Communication Coverage

- **Dead Zones:** Expect communication challenges in remote stretches between major towns; ensure up-to-date maps and alternative communication means are available.
- **Network Providers:** Major providers generally offer good coverage, but verify with local operators for specific blind spots.

## Emergency Response Times

- **Urban Areas:** Faster response times (under 30 minutes) near major towns and highways.
- **Rural Areas:** Response may be delayed (up to an hour or more) in less accessible areas; having a plan for emergency measures is essential.

## Overall Summary of Risk Assessment

The route from Sangagiri to Uddanapalli involves both urban and rural stretches requiring heightened awareness of weather conditions, road quality, and traffic patterns. Ensure compliance with all transportation regulations for hazardous materials and prepare for emergencies with backup plans and alternative routes. Finally, maintain proactive communication and risk mitigation strategies to ensure a safe and efficient journey.

### Risk Assessment - Turns

|   | Risk Type | Risk Level | Coordinates        | Speed Limit |
|---|-----------|------------|--------------------|-------------|
| 0 | Turn      | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 1 | Turn      | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 2 | Turn      | High       | 11.44900, 77.87406 | 15 KM/Hr    |
| 3 | Turn      | High       | 11.45028, 77.87202 | 15 KM/Hr    |
| 4 | Turn      | Medium     | 11.46400, 77.87323 | 30 KM/Hr    |
| 5 | Turn      | High       | 11.47403, 77.86841 | 15 KM/Hr    |
| 6 | Turn      | High       | 11.49149, 77.89601 | 15 KM/Hr    |
| 7 | Turn      | High       | 11.49224, 77.89601 | 15 KM/Hr    |
| 8 | Turn      | Medium     | 11.53512, 77.95717 | 30 KM/Hr    |
| 9 | Turn      | Medium     | 11.53606, 77.95869 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 10 | Turn       | Medium     | 11.62403, 78.10893 | 30 KM/Hr    |
| 11 | Turn       | High       | 11.64183, 78.11810 | 15 KM/Hr    |
| 12 | Turn       | Medium     | 11.64338, 78.11998 | 30 KM/Hr    |
| 13 | Turn       | Medium     | 11.73968, 78.05098 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.86790, 78.08347 | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.91357, 78.05868 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.92556, 78.06499 | 30 KM/Hr    |
| 17 | Turn       | Medium     | 11.93694, 78.05425 | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.95531, 78.04698 | 30 KM/Hr    |
| 19 | Turn       | High       | 11.95803, 78.04964 | 15 KM/Hr    |
| 20 | Turn       | Medium     | 11.96068, 78.04960 | 30 KM/Hr    |
| 21 | Turn       | High       | 11.96370, 78.05142 | 15 KM/Hr    |
| 22 | Turn       | High       | 11.96542, 78.04938 | 15 KM/Hr    |
| 23 | Turn       | High       | 11.96650, 78.04927 | 15 KM/Hr    |
| 24 | Turn       | Medium     | 12.12036, 78.13284 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 12.50466, 78.03128 | 30 KM/Hr    |
| 26 | Blind Spot | Blind Spot | 12.51373, 78.03208 | 10 KM/Hr    |
| 27 | Turn       | Medium     | 12.51382, 78.02925 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 12.51628, 78.02574 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 12.52905, 78.02044 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 12.61032, 77.93611 | 30 KM/Hr    |

Emergency Locations

|   | type     | name                    | coordinates            | speed_limit | risk_level |
|---|----------|-------------------------|------------------------|-------------|------------|
| 0 | hospital | Thilagam Hospital       | 11.4753573, 77.8685829 | 30 km/h     | Medium     |
| 1 | clinic   | Meyyazhagan Clinic      | 11.4752254, 77.8685671 | 30 km/h     | Medium     |
| 2 | hospital | Dr. Jaganathan Hospital | 11.4752717, 77.8709312 | 30 km/h     | Medium     |
| 3 | hospital | Kathir Eye Hospital     | 11.4740976, 77.8681591 | 30 km/h     | Medium     |

|    | type     | name                                                            | coordinates               | speed_limit | risk_level |
|----|----------|-----------------------------------------------------------------|---------------------------|-------------|------------|
| 4  | police   | Police Quarters                                                 | 11.4741243,<br>77.8673495 | 30 km/h     | Medium     |
| 5  | hospital | Sankari Hospital                                                | 11.4777496,<br>77.873892  | 30 km/h     | Medium     |
| 6  | clinic   | Dr. Dhanasekar Clinic                                           | 11.4793113,<br>77.8829247 | 30 km/h     | Medium     |
| 7  | hospital | Sri Vellaiyan Hospital                                          | 11.4798876,<br>77.8831394 | 30 km/h     | Medium     |
| 9  | hospital | C. N. Hospital                                                  | 11.55724,<br>78.0096      | 30 km/h     | Medium     |
| 10 | hospital | Annapoorna Medical College Hospital                             | 11.57693,<br>78.037627    | 30 km/h     | Medium     |
| 11 | hospital | Vinayaka Mission's Homeopathic Medical College & Hospital       | 11.583617,<br>78.059145   | 30 km/h     | Medium     |
| 12 | hospital | Vinayaka mission Kirubananadah variyar medical college hospital | 11.584984,<br>78.0618417  | 30 km/h     | Medium     |
| 13 | hospital | vims                                                            | 11.5854239,<br>78.0639188 | 30 km/h     | Medium     |
| 15 | hospital | Vinayaka Mission’s Kirupananda Variyar Medical College          | 11.5977669,<br>78.0850992 | 30 km/h     | Medium     |
| 20 | hospital | Bharathi Hospital, Salem                                        | 11.6492647,<br>78.1204131 | 30 km/h     | Medium     |
| 21 | hospital | Sisu Hospital Inida Ltd                                         | 11.6505276,<br>78.1205003 | 30 km/h     | Medium     |
| 22 | hospital | Aishwarya Hospital                                              | 11.6716291,<br>78.1267807 | 30 km/h     | Medium     |
| 24 | hospital | C.M. Hospital                                                   | 11.6747345,<br>78.1255034 | 30 km/h     | Medium     |
| 25 | pharmacy | Healthy Pharmacy                                                | 11.6762714,<br>78.127744  | 30 km/h     | Medium     |
| 26 | pharmacy | Thulasi Pharmacy                                                | 11.6761991,<br>78.1277377 | 30 km/h     | Medium     |
| 27 | hospital | Sri Gokulam Speciality Hospital                                 | 11.6788724,<br>78.1274258 | 30 km/h     | Medium     |
| 28 | hospital | Global Medical Center                                           | 11.6867049,<br>78.1186697 | 30 km/h     | Medium     |
| 29 | hospital | Global Medical Centre Jagirammalayam                            | 11.6894409,<br>78.1108847 | 30 km/h     | Medium     |
| 30 | hospital | Manipal Hospital                                                | 11.703072,<br>78.1026612  | 30 km/h     | Medium     |

|    | type     | name                               | coordinates               | speed_limit | risk_level |
|----|----------|------------------------------------|---------------------------|-------------|------------|
| 34 | clinic   | Manimuthu Skin Clinic              | 11.7521189,<br>78.0484058 | 30 km/h     | Medium     |
| 35 | hospital | St. Joseph's Speciality Hospital   | 11.761611,<br>78.0522344  | 30 km/h     | Medium     |
| 37 | hospital | Anugraha Multi Speciality Hospital | 11.773549,<br>78.0546408  | 30 km/h     | Medium     |
| 39 | hospital | L.R. Hospital                      | 11.8703354,<br>78.0844554 | 30 km/h     | Medium     |
| 40 | clinic   | JSA Clinic                         | 11.8700619,<br>78.0844982 | 30 km/h     | Medium     |
| 41 | hospital | Thoppur,Government Hospital        | 11.9398824,<br>78.0551658 | 30 km/h     | Medium     |

Crowded Spots

|    | type       | name                                                    | coordinates               | speed_limit | risk_level |
|----|------------|---------------------------------------------------------|---------------------------|-------------|------------|
| 14 | school     | Vinayaka Missions Annapoorani Matric School             | 11.5971662,<br>78.0839876 | 30 km/h     | Medium     |
| 16 | university | Vinayaka Missions Vinayaka Missions Research Foundation | 11.5974056,<br>78.0843617 | 30 km/h     | Medium     |
| 17 | school     | Dist Education Training                                 | 11.6133799,<br>78.1022409 | 30 km/h     | Medium     |
| 18 | school     | DIET - Salem                                            | 11.6144158,<br>78.1025078 | 30 km/h     | Medium     |
| 19 | school     | Diamond Rays Matric HR Sec School                       | 11.6385629,<br>78.1206259 | 30 km/h     | Medium     |
| 23 | college    | Sona College                                            | 11.6768,<br>78.1257052    | 30 km/h     | Medium     |
| 31 | college    | Govt Engineering College                                | 11.7128537,<br>78.0877084 | 30 km/h     | Medium     |
| 32 | university | Periyar University                                      | 11.7186758,<br>78.0771089 | 30 km/h     | Medium     |
| 33 | school     | Fathima Hr Sec School                                   | 11.7410508,<br>78.0479272 | 30 km/h     | Medium     |
| 36 | school     | John Britto Matriculation Higher Secondary School       | 11.7676907,<br>78.0513092 | 30 km/h     | Medium     |